ALLEN CAREER INSTITUTE
path to success
KOTA (RAJASTHAN)



INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Salt Analysis

English Medium

EXERCISE # I

ANIONS : Class A (Subgroup - I)

1. The colour developed, when sodium sulphide is added to sodium nitroprusside is:

(A) Purple (B) yellow (C) red (D) black

SA0001

2. When a neutral or slightly alkaline solution of thiosulphate is treated with the $[Ni(en)_3]$ $(NO_3)_2$ complex, then

(A) Green precipitate is obtained (B) Brown precipitate is obtained

(C) Violet precipitate is obtained (D) Yellow precipitate is obtained

SA0002

3. When CH_3COONa heated with solid As_2O_3 then compound X is formed. The smell of compound X is -

(A) Pungent smell (B) Rotten Fish smell (C) Nauseating smell (D) Rotten egg smell

SA0003

4. NO_2^- ion can be destroyed by -

(A) Sulphamic acid (B) Thiourea (C) Urea (D) All of these

SA0004

5. Solutions of sodium azide(NaN_3) and iodine (as KI_3) do not react but on addition of a trace of 'X' ion, which acts as a catalyst there is an immediate vigorous evolution of nitrogen. Then 'X' may be:

(A) $S_2O_3^{2-}$ (B) S^{2-} (C) $\overset{\ominus}{S}CN$ (D) All are correct.

SA0005

6. When $AgNO_3$ react with 'X' ion then initially no visible change occurs due to formation of water soluble complex.Then ion 'X' may be:

(A) SO_3^{2-} (B) $S_2O_3^{2-}$ (C) S^{2-} (D) CO_3^{2-}

SA0006

7. Match the column

Column-I	**Column-II**
(A) S^{2-}	(P) Produces white ppt. with excess $AgNO_3$
(B) HSO_3^-	(Q) Evolves gas with dil. HCl which turns lime water milky
(C) SO_3^{2-}	(R) Evolves gas with dil.H_2SO_4 which does **not** turn Baryta water milky
(D) $S_2O_3^{2-}$	(S) Produces ppt. with $Pb(OAc)_2$ solution.
	(T) Produces white ppt with $BaCl_2$ solution.

SA0007

8. Find the number of acidic radical(s) which can form coloured gas when treated with **dil.** H_2SO_4. CO_3^{2-}, NO_2^-, Br^-, I^-, SO_3^{2-}

SA0008

Class A (Subgroup - II)

9. Chromyl chloride test is given by -

(A) CH_3Cl (B) AgCl (C) Hg_2Cl_2 (D) NH_4Cl

SA0009

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10. $BO_3^{3-} + \underset{\text{Conc.}}{H_2SO_4} \xrightarrow{\Delta} \underset{\text{White fumes}}{(P)}$

$BO_3^{3-} + \underset{\text{Conc.}}{H_2SO_4} + C_2H_5OH \xrightarrow{\Delta} \underset{\text{Vapours}}{(Q)}$

P & Q are respectively -

(A) H_3BO_3 , H_3BO_3 (B) $(C_2H_5)_3BO_3$, H_3BO_3

(C) $(C_2H_5)_3BO_3$, $(C_2H_5)_3BO_3$ (D) H_3BO_3, $(C_2H_5)_3BO_3$

SA0010

11. In layer test of I^- and Br^-. If reddish -brown layer comes first then -

(A) Br^- present (B) I^- absent (C) Both (A) and (B) (D) None of these

SA0011

All Anions Of Class A

12. **Statement-1** : When H_2S gas is passed through Na-nitroprusside soluton it gives purple colouration

Statement-2 : H_2S is an weak acid

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

SA0012

13. When the soda extract containing thiosulphate ion treated with excess of $AgNO_3$ solution followed by boiling, then.

(A) White precipitate is formed (B) Black precipitate is formed

(C) brown precipitate is formed (D) No ppt precipitate is formed

SA0013

14. "Cacodyl oxide" is formed in the specific test of -

(A) Formate (B) Oxalate (C) Acetate (D) Nitrate

SA0014

15. An aqueous solution of gas (X) gives the white turbidity on passing H_2S in the solution. Identify (X)

(A) NH_3 (B) SO_2 (C) CO_2 (D) None of these

SA0015

16. NO_2^- and NO_3^- can be distinguished by which of the following reagent.

(A) dil. H_2SO_4 (B) conc. H_2SO_4

(C) Devarda's alloy + conc.NaOH (D) None of these

SA0016

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17. $[Fe(H_2O)_5NO]^{2+}$ is unstable because -

(A) It liberates NO gas on warming

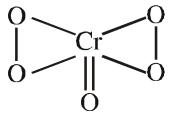
(B) It liberates NO gas on shaking

(C) The charge of central atom is +1 (relatively low enough)

(D) None of these

Class B

SA0017

18. $Cr_2O_7^{2-} + 4H_2O_2 + 2H^+ \xrightarrow[\text{Solvent}]{\text{Organic}} 2$  $+ 5H_2O$

In above reaction amyl alcohol is recommended.

Dimethyl ether is not recommended for general use owing to its -

(A) Highly non-flammable character

(B) Highly inflammable character

(C) Highly poisonious character

(D) None of these

SA0018

19. If barium sulphate is precipitated in a solution containing potassium permanganate it is coloured pink (violet) by -

(A) Absorption of some of the permanganate (B) Adsorption of some of the permanganate

(C) Both (A) and (B) (D) None of these

SA0019

All Anions Of Class A & Class B

20.

List–I (Reaction)	List–II (Product)
(P) $KI + NO_2^- \longrightarrow$	(1) NH_3
(Q) $NH_4NO_3 \xrightarrow{\Delta}$	(2) NO
(R) $NO_2^- \xrightarrow{Zn+NaOH}$	(3) N_2
(S) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$	(4) N_2O

Code :

	P	Q	R	S
(A)	3	4	1	2
(C)	4	2	3	1

	P	Q	R	S
(B)	4	2	1	3
(D)	2	4	1	3

SA0020

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21. **List–I (Reaction)** **List–II (Product)**

(P) $CO_3^{2-} \xrightarrow{dil.H_2SO_4}$ (1) H_2S

(Q) $S^{2-} \xrightarrow{dil.H_2SO_4}$ (2) SO_2

(R) $SO_3^{2-} \xrightarrow{dil.H_2SO_4}$ (3) CO_2

(S) $S_2O_3^{2-} \xrightarrow{dil.H_2SO_4}$ (4) $S + SO_2$

Code :

	P	Q	R	S		P	Q	R	S
(A)	3	4	1	2	(B)	2	1	4	3
(C)	3	1	2	4	(D)	2	4	1	3

SA0021

22. **List–I (Molecule)** **List–II (Characteristic Odour)**

(P) CO_2 (1) Rotten egg smell

(Q) SO_2 (2) Suffocating smell of burning sulphur

(R) H_2S (3) Vineger like smell

(S) CH_3COOH (4) Odour less

Code :

	P	Q	R	S		P	Q	R	S
(A)	4	2	1	3	(B)	2	4	1	3
(C)	3	1	2	4	(D)	2	4	1	3

SA0022

23. **List–I (Acidic radicals)** **List–II (Test)**

(P) NO_2^- (1) Green flame test

(Q) BO_3^{3-} (2) Cacodyl oxide reaction

(R) Br^- (3) Griess Ilosvay test

(S) CH_3COO^- (4) Layer test

Code :

	P	Q	R	S		P	Q	R	S
(A)	4	2	1	3	(B)	3	1	4	2
(C)	3	4	2	1	(D)	4	3	2	1

SA0023

CATIONS : DRY TEST

24. Find the number of water of crystallization in microcosmic salt -

(A) 5 (B) 4 (C) 6 (D) 10

SA0024

25. What is the colour of K^+ through cobalt/double blue glass -

(A) Lilac, (B) Violet (C) Brick red (D) Crimson red

SA0025

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26. What is the colour of $CoO.Al_2O_3$ is -

(A) pink (B) Thenard blue (C) Bluish white (D) None of these

SA0026

27. The correct formula of Canary yellow ppt and it is the test of ---------- acid radical-

(A) $(NH_4)_2$ $[PMo_{12}O_{40}]$ and phosphate

(B) (NH_4) H $[P(Mo_3O_{10})_4]$ and sulphate

(C) $(NH_4)_3$ $[P(Mo_3O_{10})_4]$ and phosphate

(D) Na_3 $[P(Mo_3O_{10})_4]$ and phosphate

SA0027

28. Sodium carbonate bead test generally used for .....compounds.

(A) Mn (B) Cr (C) Zn (D) Cu

SA0028

WET TEST : GROUP ZERO

29. **Statement-1 :** Test of NH_4^+ can not be done within group analysis

Statement-2 : During group analysis several times NH_4^+ - compound is added at the different steps.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is false, statement-2 is true.

(D) Statement-1 is true, statement-2 is false.

SA0029

GROUP - I

30. Which of the following is not group-I cation though the chlorides of all cations are sparingly soluble in water.

(A) Ag^+ (B) Hg_2^{2+} (C) Cu^+ (D) Pb^{2+}

SA0030

GROUP - II A

31. Cu^{2+} + KCN (in excess) $\rightarrow$ soluble complex (X). How many statements are correct regarding complex (X) -

(i) the central atom has the co-ordination number of 6

(ii) the central atom has the co-ordination number of 4

(iii) the complex is sq.planar

(iv) the complex is diamagnetic

(v) the complex is paramagnetic

SA0031

32. $BiCl_3 \xrightarrow{KI}$ black ppt (M) $\xrightarrow{\text{excess KI}}$ soluble complex (N)

Find the number of moles of I^- ions involved for the formation of per mole of (N).

SA0032

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GROUP - II B

33. Sn^{2+} and Sn^{4+} can be distinguished by how many of the following methods -

(i) by passing H_2S in their solution (in acidic medium)

(ii) by addition of NaOH in their solution

(iii) by addition of excess NaOH in their solution

(iv) by addition of dil. HCl in their solution

(v) by addition of $HgCl_2$ solution in their solution

SA0033

GROUP - III

34. What is the group-III reagent is generally used for group analysis.

(A) $NH_4OH + NH_4NO_3$ (B) $NH_4Cl + (NH_4)_2CO_3$

(C) $NH_4OH + (NH_4)_2SO_4$ (D) $NH_4OH + NH_4Cl$

SA0034

35. $CrCl_3$ solution + Na_2S solution $\longrightarrow$ ppt(A)

The correct formula and colour of A are

(A) Cr_2S_3, Black (B) $Cr(OH)_3$, Green

(C) $Na[Cr(OH)_4]$, Green (D) None of these

SA0035

GROUP - IV

36. The auxiliary reagent in group-IV reagent is

(A) H_2S (B) dil.HCl (C) NaOH (D) NH_4OH

SA0036

All Group Cations

37. Which of the following cation gives ppt in two groups during group analysis.

(A) Hg^{2+} (B) Hg_2^{2+} (C) Pb^{2+} (D) Cu^{2+}

SA0037

38. Which of the following cation produces coloured ppt with Na_2SO_4 solution -

(A) Pb^{2+} solution (B) Ba^{2+} solution (C) Hg^{2+} solution (D) Ca^{2+} solution

SA0038

39. NH_4^+ and K^+ ions can be distinguished by the use of following reagent

(A) $Na_3[Co(NO_2)_6]$ (B) $Na_2[PtCl_6]$

(C) $HClO_4$ or $NaClO_4$ (D) Boiling with NaOH

SA0039

40. Which of the following sulphides is yellow in colour?

(A) CuS (B) CdS (C) ZnS (D) CoS

SA0040

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MISCELLANEOUS

41.

List–I (Compound)

(P) HgO

(Q) $BaCO_3$

(R) $Na_4[Fe(CN)_5NOS]$

(S) KI_3

List–II (Colour)

(1) Purple solution

(2) Yellow ppt

(3) Dark brown

(4) White ppt

Code :	**P**	**Q**	**R**	**S**
(A)	3	4	1	2
(B)	2	4	1	3
(C)	2	4	3	1
(D)	2	4	3	1

SA0041

42.

List–I (Basic Radical)

(P) Al^{+3}

(Q) Zn^{+2}

(R) Ba^{+2}

(S) Pb^{+2}

List–II (Group)

(1) II group

(2) V group

(3) IV group

(4) III group

Code :	**P**	**Q**	**R**	**S**
(A)	4	2	1	3
(B)	2	4	1	3
(C)	3	1	2	4
(D)	4	3	2	1

SA0042

43.

List–I (Cations)

(P) Co^{+2}

(Q) Fe^{+3}

(R) Cu^{+2}

(S) Ca^{+2}

List–II (Group reagent)

(1) $(NH_4)_2CO_3$ in presence of NH_4Cl

(2) H_2S gas in acidic medium

(3) H_2S in presence of NH_4OH

(4) NH_4OH in presence of NH_4Cl

Code :	**P**	**Q**	**R**	**S**
(A)	4	2	1	3
(B)	3	1	4	2
(C)	1	2	3	4
(D)	3	4	2	1

SA0043

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EXERCISE # II

ANIONS : Class A (Subgroup - I)

1. **Statement-1:** On passing CO_2 gas through lime water, the solution turns milky.
because
Statement-2: Acid-Base (neutralisation) reaction takes place.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

SA0044

2. A substance on treatment with dil. H_2SO_4 liberates a colourless gas which produces (i) turbidity with baryta water and (ii) turns acidified dichromate solution green. The reaction indicates the presence of

(A) CO_3^{2-} (B) S^{2-} (C) SO_3^{2-} (D) NO_2^-

SA0045

3. When $S_2O_3^{2-}$ react with solution of 'X' reagent then reaction is redox followed by precipitation then 'X' is:
(A) $FeCl_3$ solution (B) $AgNO_3$ solution
(C) $CuSO_4$ solution (D) None of these

SA0046

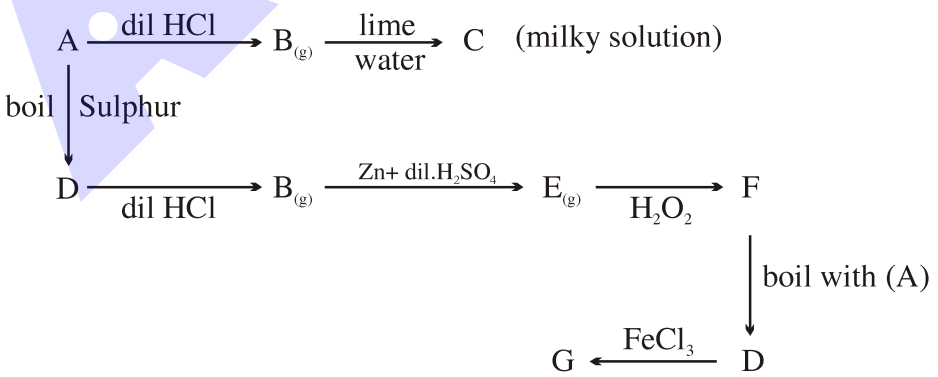
4. In the test for iodine, when I_2 is treated with sodium thiosulphate, $Na_2S_2O_3$

$$Na_2S_2O_3 + I_2 \longrightarrow NaI + ...$$

(A) $Na_2S_4O_6$ (B) Na_2SO_4 (C) Na_2S (D) Na_3ISO_4

SA0047

Paragraph for Q. 5 to Q. 7



5. Identify A
(A) CO_3^{2-} (B) SO_3^{2-} (C) $S_2O_3^{2-}$ (D) none of these

SA0048

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6. When A react with $Pb(NO_3)_2$ then compound X is formed. Compound X is oxidized by atmospheric oxygen on boiling, then Y is formed what is the colour of Y

(A) yellow (B) White (C) Black (D) Green

SA0048

7. When gas E react with sodium nitroprusside in basic medium then compound Z is formed. The colour of compound Z is:

(A) Green (B) purple (C) Reddish brown (D) Black

SA0048

Class A (Subgroup - II)

8. When a mixture of solid NaCl, solid $K_2Cr_2O_7$ is heated with conc. H_2SO_4, orange red vapours are obtained. These are of the compound

(A) chromous chloride (B) chromyl chloride

(C) chromic chloride (D) chromic sulphate

SA0049

9. Which of the following will not give positive chromyl chloride test?

(A) Copper chloride, $CuCl_2$ (B) Mercuric chloride, $HgCl_2$

(C) Zinc chloride, $ZnCl_2$ (D) Aniline chloride, $C_6H_5NH_3Cl$

SA0050

10. Sodium borate on reaction with conc. H_2SO_4 and C_2H_5OH gives a compound A which burns with a green edged flame. The compound A is

(A) $H_2B_4O_7$ (B) $(C_2H_5)_2B_4O_7$ (C) H_3BO_3 (D) $(C_2H_5)_3BO_3$

SA0051

11. Nitrate is confirmed by ring test. The brown colour of the ring is due to formation of

(A) ferrous nitrite (B) nitroso ferrous sulphate

(C) ferrous nitrate (D) $FeSO_4NO_2$

SA0052

12. A salt gives violet vapours when treated with conc. H_2SO_4, it contains

(A) Cl^- (B) I^- (C) Br^- (D) NO_3^-

SA0053

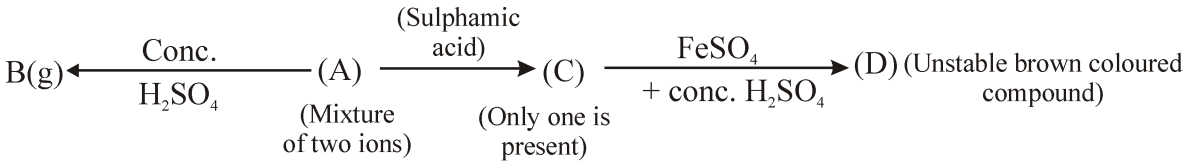
13. Unknown salt + Al-powder + NaOH (conc.) → gas comes out which turns Nesslar's reagent brown. The salt may be -

(A) $NaNO_2$ (B) $NaNO_3$ (C) NH_4Cl (D) NH_4HCO_3

SA0054

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Paragraph for Q. 14 to Q. 17



14. Identify mixture of ions (A) -

(A) NO_2^- and Br^- (B) NO_2^- and I^- (C) NO_2^- and NO_3^- (D) None of these

SA0055

15. What is oxidation state of central atom of (D)

(A) +3 (B) +2 (C) +1 (D) Zero

SA0055

16. Identify gas B-

(A) Br_2 (B) $Br_2 + NO_2$ (C) NO_2 (D) None of these

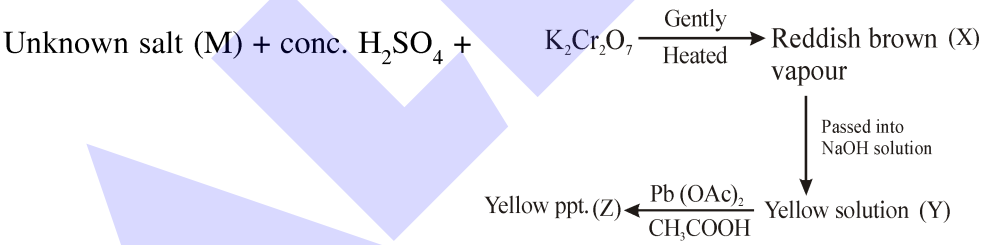
SA0055

17. What is the hybridisation of central atom of D-

(A) d^2sp^3 (B) sp^3d^2 (C) sp^3d (D) sp^3

SA0055

Paragraph for Q. 18 to Q. 20



18. The salt (M) is/are-

(A) AgCl (B) NH_4Cl + NaBr (C) NaBr (D) $Ca(ClO_4)_2$

SA0056

19. How many non axial d-orbitals are involved in hybridisation of central atom of compound (X)-

(A) 2 (B) 3 (C) 4 (D) None of these

SA0056

20. What is the formula of yellow ppt (Z)-

(A) $BaCrO_4$ (B) Na_2CrO_4 (C) Ag_2CrO_4 (D) $PbCrO_4$

SA0056

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CATIONS : DRY TEST

21. In the borax bead test of Co^{2+}, the blue colour of bead is due to the formation of:
(A) B_2O_3 (B) Co_3B_2 (C) $Co(BO_2)_2$ (D) CoO

SA0057

22. Which of the following leaves no residue on heating?
(A) $Pb(NO_3)_2$ (B) NH_4NO_3 (C) $Cu(NO_3)_2$ (D) $NaNO_3$

SA0058

23. Which of the following cations is detected by the flame test?
(A) NH_4^+ (B) K^+ (C) Mg^{2+} (D) Al^{3+}

SA0059

24. Which metal salt gives a violet coloured bead in the borax bead test (oxidising flame, cold)?
(A) Fe^{2+} (B) Ni^{2+} (C) Co^{2+} (D) Mn^{2+}

SA0060

25. The compound formed in the borax bead test of Cu^{2+} ion in oxidising flame is:
(A) Cu (B) $CuBO_2$ (C) $Cu(BO_2)_2$ (D) None of these

SA0061

26. In microcosmic salt bead test Co^{2+} produce blue bead due to the formation of -
(A) $Cu(BO_2)_2$ (B) $NaCoPO_4$ (C) $Co_2(PO_4)(BO_2)$ (D) $NaPO_3$

SA0062

Paragraph for Q. 27 to Q. 30

$$\underset{\text{(Hydrated salt)}}{(A)} \xrightarrow[\text{heated}]{\text{Strongly}} \underbrace{B + C}_{\substack{\text{transparent}\\ \text{glassy bead}}}$$

$$H_3BO_3 \xrightarrow[\text{heated}]{\text{Strongly}} C + D$$

27. Identify C-
(A) $(BN)_X$ (B) $NaPO_3$ (C) B_2O_3 (D) $Mg(NH_4)PO_4$

SA0063

28. Find the number of water of crystallizations in (A)-
(A) 4 (B) 5 (C) 10 (D) 24

SA0063

29. How many X–O–X linkages are present in structure of A (X = central atom)-
(A) 4 (B) 3 (C) 5 (D) 2

SA0063

30. Find the number of tetrahedral and trigonal planar units in structure of A -
(A) 2,1 (B) 2,2 (C) 2,4 (D) 5,2

SA0063

WET TEST : GROUP - I

31. Mercurous ion is represented as:
(A) Hg_2^{2+} (B) Hg^{2+} (C) $Hg + Hg^{2+}$ (D) Hg_2^+

SA0064

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32. A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the solution, a white precipitate is obtained which does not dissolve in dil. HNO_3. The anion could be:

(A) CO_3^{2-} (B) Cl^- (C) SO_4^{2-} (D) S^{2-}

SA0065

33. An aqueous solution of a substance gives a white ppt. on treatment with dil. HCl, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black ppt. is obtained. The substance is a

(A) Hg^{2+} salt (B) Cu^{2+} salt (C) Ag^+ salt (D) Pb^{2+} salt

SA0066

34. A white ppt obtained in a analysis of a mixture becomes black on treatment with NH_4OH. It may be

(A) $PbCl_2$ (B) AgCl (C) $HgCl_2$ (D) Hg_2Cl_2

SA0067

GROUP - II

35. When bismuth chloride is poured into a large volume of water the white precipitate produced is

(A) $Bi(OH)_3$ (B) Bi_2O_3 (C) BiOCl (D) Bi_2OCl_3

SA0068

36. $CuSO_4$ decolourises on addition of excess KCN, the product is

(A) $[Cu(CN)_4]^{2-}$.

(B) Cu^{2+} get reduced to form $[Cu(CN)_4]^{3-}$

(C) $Cu(CN)_2$

(D) CuCN

SA0069

37. When H_2S gas is passed through the HCl containing aqueous solution of $CuCl_2$, $HgCl_2$, $BiCl_3$ and $CoCl_2$, it does not precipitate out:

(A) CuS (B) HgS (C) Bi_2S_3 (D) CoS

SA0070

38. Which of the following is soluble in yellow ammonium sulphide?

(A) CuS (B) CdS (C) SnS (D) PbS

SA0071

39. When excess of $SnCl_2$ is added to a solution of $HgCl_2$, a white ppt turning grey is obtained. The grey colour is due to the formation of

(A) Hg_2Cl_2 (B) $SnCl_4$ (C) Sn (D) Hg

SA0072

40. On passing H_2S gas in II group sometimes the solution turns milky. It indicates the presence of

(A) oxidising agent (B) acidic salt (C) s-block cation (D) reducing agent.

SA0073

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\6-Salt Anal-Heating Effs,d-Block-Ran of Salt Anal-(OC)\Eng\01_SA\01_ThEx.p65

41. Which fo the following yellow coloured sulphide is insouble in yellow ammonium sulphide.

(A) SnS_2 (B) As_2S_5 (C) CdS (D) Bi_2S_3

SA0074

42. Type of sulphide ppt may be obtained in the group-II ppt during group analysis.

(A) M_2S_3 (B) M_2S (C) MS (D) MS_2

SA0075

43. The metal ion which is precipitated when H_2S is passed with HCl:

(A) Zn^{2+} (B) Ni^{2+} (C) Cd^{2+} (D) Mn^{2+}

SA0076

GROUP - III

44. In the precipitation of the iron group in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to

(A) decrease concentration of OH^- ions. (B) prevent interference by phosphate ions.

(C) increase concentration of Cl^- ions. (D) increase concentration of NH_4^+ ions.

SA0077

45. If reddish brown ppt (only) is obtained in group-III during group analysis, then oxidation state of Fe in the original sample may be

(A) +2 (B) +3 (C) +2 and +3 both (D) Neither +2 nor +3

SA0078

46. If NH_4Cl is not added to the group-III reagent which of the following ppt could be obtained

(A) $Cr(OH)_2$ (B) $Fe(OH)_3$ (C) $Mn(OH)_2$ (D) $Mg(OH)_2$

SA0079

47. In which of the following cases blue ppt is obtained

(A) $Fe^{2+} + [Fe(CN)_6]^{3-} \longrightarrow$ (B) $Fe^{2+} + [Fe(CN)_6]^{4-} \longrightarrow$

(C) $Fe^{3+} + [Fe(CN)_6]^{4-} \longrightarrow$ (D) $Fe^{3+} + [Fe(CN)_6]^{3-} \xrightarrow{SnCl_2}$

SA0080

48. What are the following steps are to be done before adding group-III reagent into the group-II filtrate.

(A) Group-II filtrate is to be evapourated to dryness

(B) Group-II filtrate is to be boiled of first

(C) After boiling 2-3 drops of dil.H_2SO_4 is added and boiled again.

(D) After boiling 2-3 drops of HNO_3 is added and boiled again.

SA0081

49. A pale green crystalline metal salt of M dissolves freely in water. On standing it gives a brown ppt on addition of aqueous NaOH. The metal salt solution also gives a black ppt on bubbling H_2S in basic medium. An aqueous solution of the metal salt decolourizes the pink colour of the permanganate solution. The metal in the metal salt solution is

(A) copper (B) aluminium (C) lead (D) iron

SA0082

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal+Heating Eff+s_d Block-Ron of Salt Anal-(OC)\Eng\01_SA\01_TheEx.p65

50. Which of the following compound on reaction with NaOH and Na_2O_2 gives yellow colour?

(A) $Cr(OH)_3$ (B) $Zn(OH)_2$ (C) $Al(OH)_3$ (D) None of these

SA0083

GROUP - IV

51. Colour of nickel chloride solution is

(A) pink (B) black (C) colourless (D) green

SA0084

52. Dimethyl glyoxime in a suitable solvent was refluxed for 10 minutes with pure pieces of nickel sheet, it will result in

(A) Red ppt (B) Blue ppt. (C) Yellow ppt. (D) No ppt.

SA0085

53. Which one of the following does not produce metallic sulphide with H_2S?

(A) $ZnCl_{2(\text{Neutral sol}^n)}$ (B) $CdCl_{2(aq)}$ (C) $CoCl_{2(aq)}$ (D) $CuCl_{2(aq)}$

SA0086

54. Which is not dissolved by dil HCl?

(A) ZnS (B) MnS (C) $BaSO_3$ (D) $BaSO_4$

SA0087

GROUP - V

55. In III group, NH_4Cl is added to decrease concentation of hydroxide ion by NH_4OH. We do not add $(NH_4)_2SO_4$ along with NH_4OH because -

(A) $(NH_4)_2SO_4$ is insoluble in water (B) It precipitate other insoluble sulphates

(C) It is weak electrolyte (D) None of these

SA0088

GROUP - VI

56. A metal is burnt in air and the ash on moistening smells of ammonia. The metal is

(A) Na (B) Fe (C) Mg (D) Al

SA0089

57. A metal 'X' on heating in nitrogen gas gives 'Y' . 'Y' on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour Y is:

(A) $Mg(NO_3)_2$ (B) Mg_3N_2 (C) NH_3 (D) MgO

SA0090

MISCELLANEOUS

58. Na_2HPO_4 + Reagent 'M' $\rightarrow$ white ppt. The reagent 'M' is -

(A) $BaCl_2$ solution (B) $AlCl_3$ solution (C) $MnSO_4$ solution (D) $FeCl_3$ solution

SA0091

59. A white solid is first heated with dil H_2SO_4 and then with conc. H_2SO_4. No action was observed in either case. The solid salt contains

(A) sulphide (B) sulphite (C) thiosulphate (D) sulphate

SA0092

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60. A mixture of chlorides of copper, cadmium, chromium, iron and aluminium was dissolved in water acidified with HCl and hydrogen sulphide gas was passed for sufficient time. It was filtered, boiled and a few drops of nitric acid were added while boiling. To this solution ammonium chloride and sodium hydroxide were added in excess and filtered. The filtrate shall give test for

(A) sodium and iron ion
(B) sodium, chromium and aluminium ion
(C) aluminium and iron ion
(D) sodium, iron, cadmium and aluminium ion

SA0093

61. In Nessler's reagent, the ion present is:

(A) HgI^{2-} (B) HgI_4^{2-} (C) Hg^{+} (D) Hg^{2+}

SA0094

62. The cations present in slightly acidic solution are Fe^{3+}, Zn^{2+} and Cu^{2+}. The reagent which when added in excess to this solution would identify and separte Fe^{3+} in one step is:

(A) 2 M HCl (B) 6 M NH_3 (C) 6 M NaOH (D) H_2S gas

SA0095

63. In the separation of Cu^{2+} and Cd^{2+} in 2nd group qualitative analysis of cation, tetrammine copper (II) sulphate and tetrammine cadmium (II) sulphate react with KCN to form the corresponding cyano complexes. Which one of the following pairs of the complexes and their relative stability enables the separation of Cu^{2+} and Cd^{2+}?

(A) $K_3[Cu(CN)_4]$ more stable and $K_2[Cd(CN)_4]$ less stable.
(B) $K_2[Cu(CN)_4]$ less stable and $K_2[Cd(CN)_4]$ more stable.
(C) $K_2[Cu(CN)_4]$ more stable and $K_2[Cd(CN)_4]$ less stable.
(D) $K_3[Cu(CN)_4]$ less stable and $K_2[Cd(CN)_4]$ more stable.

SA0096

64. Which one has the minimum solubility product?

(A) AgCl (B) $AlCl_3$ (C) $BaCl_2$ (D) NH_4Cl

SA0097

65. Which of the following sulphate is insoluble in water?

(A) $CuSO_4$ (B) $CdSO_4$ (C) $PbSO_4$ (D) $Bi_2(SO_4)_3$

SA0098

66. Which of the following gives blood red colour with KSCN?

(A) Cu^{2+} (B) Fe^{3+} (C) Al^{3+} (D) Zn^{2+}

SA0099

67. Which one of the following metal sulphide has maximum solubility in water?

(A) HgS, $K_{sp} = 10^{-54}$
(B) CdS, $K_{sp} = 10^{-30}$
(C) FeS, $K_{sp} = 10^{-20}$
(D) ZnS, $K_{sp} = 10^{-22}$

SA0100

node06\B0BC-BD-Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal+Heating Eff+s_d Block-Ran of Salt Anal-(OC)\Eng\01_SA\01_ThrEx.p65

68. Identify the correct order of solubility of Na_2S, CuS and ZnS in aqueous medium is:

(A) CuS > ZnS > Na_2S (B) ZnS > Na_2S >CuS

(C) Na_2S > CuS > ZnS (D) Na_2S > ZnS > CuS

SA0101

69. Match the column -

Column-I (Element)	**Column-II (Correct characteristics)**
(A) Ba	(P) cation in solution produces brick red ppt. with CrO_4^{2-}
(B) Pb	(Q) cation in solution produces yellow ppt .with CrO_4^{2-}
(C) Ag	(R) corresponding salt produces apple green colour in the flame test
(D) Ca	(S) corresponding salt produces brick red colour in the flame test
	(T) cation in solution produces no ppt. with CrO_4^{2-} ion

SA0102

70.

Column-I Cation in solution	**Column-II Correct characteristics when no where excess reagent is used**
(A) Ag^+ and Pb^{2+}	(P) can be distinguished by Na_2HPO_4 solution
(B) Zn^{2+} and Mg^{2+}	(Q) can be distinguished by dil.HCl
(C) Pb^{2+} and Hg_2^{2+}	(R) can be distinguished by KI solution
(D) Ag^+ and Fe^{3+}	(S) can not be distinguished by NH_4OH solution

SA0103

The following column 1, 2, 3 represent the various tests carried out for identification of various group basic radicals, using various reagents and nature of reaction/properties of products observed. Answer the questions that follow

Column-1 - Cations/Basic Radical

Column-2 - Excess Reagent used with cation

Column-3 - Nature of Reaction/Properties of product formed

Column - 1 Cations	**Column - 2 Excess Reagent used with cation**	**Column - 3 Nature of Reaction/ Properties of product formed**
(I) Cu^{2+}	(i) KI (< 6M)	(P) Reduction of cation occurs
(II) Fe^{3+}	(ii) $K_4[Fe(CN)_6]$	(Q) Coloured complex formation
(III) Pb^{2+}	(iii) KCN	(R) Precipitation occurs
(IV) Ni^{2+}	(iv) NH_4OH	(S) Diamagnetic & square planar complex formation

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71. For a **group-II** basic radical, which is the only **INCORRECT** combination?

(A) (I), (i), (P) (B) (IV), (iii), (S) (C) (III), (iv), (R) (D) (III), (iii), (R)

SA0104

72. For a **group-IV** basic radical, which is the only **CORRECT** combination?

(A) (I), (iv), (S) (B) (IV), (iii), (P) (C) (II), (iv), (Q) (D) (IV), (iv), (Q)

SA0104

73. Which combination has a entirely different colour from others?

(A) (IV), (iv), (Q) (B) (I), (iv), (Q) (C) (II), (iii), (Q) (D) (II), (ii), (Q)

SA0104

74. How many of the following gives green ppt.

(i) $CrCl_3$ + NaOH $\rightarrow$ (ii) $CrCl_3$ + excess NaOH $\rightarrow$

(iii) $NiCl_2$ + excess NaOH $\rightarrow$ (iv) $NiCl_2$ + excess NH_4OH $\rightarrow$

(v) Hg_2^{2+} + KI $\rightarrow$

SA0104

75. Find the no. of cation which gives white ppt with $K_4[Fe(CN)_6]$

Sr^{2+} Ca^{2+}, Zn^{2+}, Fe^{3+} , Cu^{2+}

SA0104

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EXERCISE # JEE MAINS

1. Which products are expected from the disproportionation of hypochlorous acid :**[AIEEE-2002]**

(1) $HClO_3$ and Cl_2O (2) $HClO_2$ and $HClO$ (3) HCl and Cl_2O (4) HCl and $HClO_3$

SA0105

2. A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH. The metal M is:-

[AIEEE-2002]

(1) Mg (2) Ba (3) Ca (4) Be

SA0106

3. Which statement is correct :- **[AIEEE-2003]**

(1) Fe^{3+} ions give deep green precipitate with $K_4[Fe(CN)_6]$

(2) On heating K^+, Ca^{2+} and HCO_3^- ions, we get a precipitate of $K_2[Ca(CO_3)_2]$

(3) Manganese salts give a violet borax bead test in reducing flame

(4) From a mixed precipitate of AgCl and AgI ammonia solution dissolves only AgCl

SA0107

4. What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid - **[AIEEE-2003]**

(1) Cr^{3+} and $Cr_2O_7^{2-}$ are formed (2) $Cr_2O_7^{2-}$ and H_2O are formed

(3) $Cr_2O_7^{2-}$ is reduced to +3 state of Cr (4) $Cr_2O_7^{2-}$ is oxidised to +7 state of Cr

SA0108

5. Ammonia forms the complex ion $[Cu(NH_3)_4]^{2+}$ with copper ions in alkaline solutions but not in acidic solution. What is the reason for it :- **[AIEEE-2003]**

(1) In acidic solutions hydration protects copper ions

(2) In acidic solutions protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available

(3) In alkaline solutions insoluble $Cu(OH)_2$ is precipitated which is soluble in excess of any alkali

(4) Copper hydroxide is an amphoteric substance

SA0109

6. Excess of KI reacts with $CuSO_4$ solution and then $Na_2S_2O_3$ solution is added to it. Which of the statements is incorrect for this reaction : **[AIEEE-2004]**

(1) Evolved I_2 is reduced (2) CuI_2 is formed

(3) $Na_2S_2O_3$ is oxidised (4) Cu_2I_2 is formed

SA0110

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7. Calomel on reaction with NH_4OH gives **[AIEEE-2004]**

(1) $HgNH_2Cl$ (2) $NH_2–Hg–Hg–Cl$ (3) Hg_2O (4) HgO

SA0111

8. One mole of magnesium nitride on reaction with excess of water gives :- **[AIEEE-2004]**

(1) Two mole of HNO_3 (2) Two mole of NH_3

(3) 1 mole of NH_3 (4) 1 mole of HNO_3

SA0112

9. The products obtained on heating $LiNO_3$ will be :- **[AIEEE-2011]**

(1) $LiNO_2 + O_2$ (2) $Li_2O + NO_2 + O_2$

(3) $Li_3N + O_2$ (4) $Li_2O + NO + O_2$

SA0113

10. What is the best description of the change that occurs when $Na_2O(s)$ is dissolved in water ?

(1) Oxidation number of sodium decreases **[AIEEE-2011]**

(2) Oxide ion accepts sharing in a pair of electrons

(3) Oxide ion donates a pair of electrons

(4) Oxidation number of oxygen increases

SA0114

11. Which of the following on thermal-decomposition yields a basic as well as an acidic oxide ? **[AIEEE-2012]**

(1) NH_4NO_3 (2) $NaNO_3$ (3) $KClO_3$ (4) $CaCO_3$

SA0115

12. The correct statement for the molecule, CsI_3, is: **[JEE(Main)-2014]**

(1) it contains Cs^{3+} and I^- ions (2) it contains Cs^+, I^- and lattice I_2 molecule

(3) it is a covalent molecule (4) it contains Cs^+ and I_3^- ions

SA0116

13. A metal M on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour, Y is :-

[JEE(Main)-2012 online_P-4]

(1) NH_3 (2) MgO (3) Mg_3N_2 (4) $Mg(NO_3)_2$

SA0117

14. Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown - red vapours of: **[JEE(Main)-2013 online_P-1]**

(1) CrO_3 (2) Cr_2O_3 (3) $CrCl_3$ (4) CrO_2Cl_2

SA0118

15. Sodium Carbonate cannot be used in place of $(NH_4)_2CO_3$ for the identification of Ca^{2+} , Ba^{2+} and Sr^{2+} ions (in group V) during mixture analysis because : **[JEE(Main)-2013 online_P-1]**

(1) Sodium ions will react with acid radicals

(2) Concentration of CO_3^{2-} ions is very low

(3) Mg^{2+} ions will also be precipitated

(4) Na^+ ions will interfere with the detection of Ca^{2+}, Ba^{2+}, Sr^{2+} ions

SA0119

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal+Heating Eff+s,d Block-Ran of Salt Anal\(OC)\Eng\01_SA\01_TH+Ex.p65

16. Which of the following statements is incorrect? **[JEE(Main)-2013 online_P-2]**

(1) Fe^{2+} ion also gives blood red colour with SCN^- ion

(2) Cupric ion reacts with excess of ammonia solution to give deep blue colour of $[Cu(NH_3)_4]^{2+}$ ion.

(3) Fe^{3+} ion gives blood red colour with SCN^- ion.

(4) On passing H_2S into Na_2ZnO_2 solution, a white ppt of ZnS is formed.

SA0120

17. Identify incorrect statement **[JEE(Main)-2013 online_P-3]**

(1) Copper (I) compounds are colourless except where colour results from charge transfer

(2) Copper (I) compounds are diamagnetic

(3) Cu_2S is black

(4) Cu_2O is colourless

SA0121

18. Which one of the following cannot function as an oxidising agent ?**[JEE(Main)-2013 online_P-4]**

(1) $NO_3^-(aq)$ (2) I^- (3) $Cr_2O_7^{2-}$ (4) $S_{(S)}$

SA0122

19. Which of the following statements about Na_2O_2 is **not** correct ? **[JEE(Main)-2014 online_P-2]**

(1) Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium

(2) It is diamagnetic in nature

(3) It is the super oxide of sodium

(4) It is a derivative of H_2O_2

SA0123

20. Consider the following equilibrium **[JEE(Main)-2014 online_P-2]**

$$AgCl\downarrow + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ + Cl^-$$

White precipitate of AgCl appears on adding which of the following?

(1) NH_3 (2) Aqueous NaCl (3) Aqueous NH_4Cl (4) Aqueous HNO_3

SA0124

21. Consider the reaction **[JEE(Main)-2014 online_P-4]**

$$H_2SO_{3(aq)} + Sn^{4+}_{(aq)} + H_2O_{(l)} \rightarrow Sn^{2+}_{(aq)} + HSO^-_{4(aq)} + 3H^+_{(aq)}$$

Which of the following statements is correct?

(1) H_2SO_3 is the reducing agent because it undergoes oxidation

(2) H_2SO_3 is the reducing agent because it undergoes reduction

(3) Sn^{4+} is the reducing agent because it undergoes oxidation

(4) Sn^{4+} is the oxidizing agent because it undergoes oxidation

SA0125

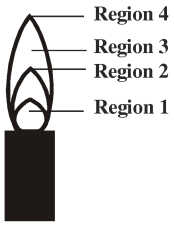
22. Amongst the following, identify the species with an atom in +6 oxidation state : **[JEE(Main)-2014 online_P-4]**

(1) $[MnO_4]^-$ (2) $[Cr(CN)_6]^{3-}$ (3) Cr_2O_3 (4) CrO_2Cl_2

SA0126

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23. The hottest region of Bunsen flame shown in the figure below is : **[JEE(Main)-2016]**



(1) region 4
(2) region 1
(3) region 2
(4) region 3

SA0127

24. Sodium extract is heated with concentrated HNO_3 before testing for halogens because :
(1) S^{2-} and CN^-, if present are decomposed by conc. HNO_3 and hence do not interfere in the test.
(2) Ag reacts faster with halides in acidic medium **[JEE(Main)-2016 online]**
(3) Ag_2S and AgCN are soluble in acidic medium.
(4) Silver halides are totally insoluble in nitric acid.

SA0128

25. In the following reactions, ZnO is respectively acting as a/an :
(a) $ZnO + Na_2O \rightarrow Na_2ZnO_2$ (b) $ZnO + CO_2 \rightarrow ZnCO_3$ **[JEE(Main)-2017 offline]**
(1) base and acid (2) base and base (3) acid and acid (4) acid and base

SA0129

26. The products obtained when chlorine gas reacts with cold and dilute aqueous NaOH are :-
(1) ClO^- and ClO_3^- (2) ClO_2^- and ClO_3^- **[JEE(Main)-2017 offline]**
(3) Cl^- and ClO^- (4) Cl^- and ClO_2^-

SA0130

27. Sodium salt of an organic acid 'X' produces effervescence with conc. H_2SO_4. 'X' reacts with the acidified aqueous $CaCl_2$ solution to give a white precipitate which decolourises acidic solution of $KMnO_4$. 'X' is :- **[JEE(Main)-2017 offline]**
(1) C_6H_5COONa (2) HCOONa (3) CH_3COONa (4) $Na_2C_2O_4$

SA0131

28. A solution containing a group-IV cation gives a precipitate on passing H_2S. A solution of this precipitate in dil. HCl produces a white precipitate with NaOH solution and bluish-white precipitate with basic potassium ferrocyanide. The cation is : **[JEE(Main)-2017 online]**
(1) Mn^{2+} (2) Zn^{2+} (3) Ni^{2+} (4) Co^{2+}

SA0132

29. Which of the following ions does **not** liberate hydrogen gas on reaction with dilute acids?
(1) Ti^{2+} (2) Cr^{2+} **[JEE(Main)-2017 online]**
(3) Mn^{2+} (4) V^{2+}

SA0133

30. When metal 'M' is treated with NaOH, a white gelatinous precipitate 'X' is obtained, which is soluble in excess of NaOH. Compound 'X' when heated strongly gives an oxide which is used in chromatography as an adsorbent. The metal 'M' is **[JEE(Main)-2018 offline]**
(1) Ca (2) Al (3) Fe (4) Zn

SA0134

31. A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the aforementioned solution, a white precipitate is obtained which does not dissolve in dil. nitric acid. The anion is : **[JEE(Main)-2018 online]**
(1) S^{2-} (2) SO_4^{2-} (3) CO_3^{2-} (4) Cl^-

SA0135

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32. The incorrect statement is :- **[JEE(Main)-2018 online]**

(1) Ferric ion gives blood red colour with potassium thiocyanate.

(2) Cu^{2+} and Ni^{2+} ions give black precipitate with H_2S in presence of HCl solution.

(3) Cu^{2+} salts give red coloured borax bead test in reducing flame.

(4) Cu^{2+} ion gives chocolate coloured precipitate with potassium ferrocyanide solution.

SA0136

33. When XO_2 is fused with an alkali metal hydroxide in presence of an oxidizing agent such as KNO_3; a dark green product is formed which disproportionates in acidic solution to afford a dark purple solution. X is : **[JEE(Main)-2018 online]**

(1) Ti (2) Cr (3) V (4) Mn

SA0137

34. Reaction of an inorganic sulphite X with dilute H_2SO_4 generates compound Y. Reaction of Y with NaOH gives X. Further, the reaction of X with Y and water affords compound Z. Y and Z, respectively, are: **[JEE MAINS 2020]**

(1) S and Na_2SO_3 (2) SO_2 and $NaHSO_3$

(3) SO_3 and $NaHSO_3$ (4) SO_2 and Na_2SO_3

SA0176

35. The reaction of white phosphorus on boiling with alkali in inert atmosphere resulted in the formation of product 'A'. The reaction 1 mol of 'A' with excess of $AgNO_3$ in aqueous medium gives _____ mol(s) of Ag. (Round off to the Nearest Integer). **[JEE MAINS 2021]**

SA0177

36. An inorganic Compound 'X' on treatment with concentrated H_2SO_4 produces brown fumes and gives dark brown ring with $FeSO_4$ in presence of concentrated H_2SO_4. Also Compound 'X' gives precipitate 'Y', when its solution in dilute HCl is treated with H_2S gas. The precipitate 'Y' on treatment with concentrated HNO_3 followed by excess of NH_4OH further gives deep blue coloured solution, Compound 'X' is: **[JEE MAINS 2021]**

(1) $Co(NO_3)_2$ (2) $Pb(NO_2)_2$ (3) $Cu(NO_3)_2$ (4) $Pb(NO_3)_2$

SA0178

37. Given below are two statements :

Statement I : Colourless cupric metaborate is reduced to cuprous metaborate in a luminous flame.

Statement II : Cuprous metaborate is obtained by heating boric anhydride and copper sulphate in a non-luminous flame.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Statement I is true but Statement II is false **[JEE MAINS 2021]**

(2) Both Statement I and Statement II are false

(3) Statement I is false but Statement II is true

(4) Both Statement I and Statement II are true

SA0179

38. White precipitate of AgCl dissolves in aqueous ammonia solution due to formation of : **SA0184**

(A) $[Ag(NH_3)_4]Cl_2$ (B) $[Ag(Cl)_2(NH_3)_2]$ **[JEE MAINS 2022]**

(C) $[Ag(NH_3)_2]Cl$ (D) $[Ag(NH_3)Cl]Cl$

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Salt Anal-Heating Effs,d-Block-Ron of Salt Anal-(OCI)\Eng\01_SA\01_ThEx.p65

39. During the qualitative analysis of salt with cation y^{2+} , addition of a reagent (X) to alkaline solution of the salt gives a bright red precipitate. The reagent (X) and the cation (y^{2+}) present respectively are:

(A) Dimethylglyoxime and Ni^{2+} **[JEE MAINS 2022]**

(B) Dimethylglyoxime and Co^{2+} **SA0185**

(C) Nessler's reagent and Hg^{2+}

(D) Nessler's reagent and Ni^{2+}

40. In the flame test of a mixture of salts, a green flame with blue centre was observed. Which one of the following cations may be present? **[JEE MAINS 2022]**

(A) Cu^{2+} (B) Sr^{2+} (C) Ba^{2+} (D) Ca^{2+} **SA0186**

41. A white precipitate was formed when $BaCl_2$ was added to water extract of an inorganic salt. Further, a gas 'X' with characteristic odour was released when the formed white precipitate was dissolved in dilute HCl. The anion present in the inorganic salt is : **[JEE MAINS 2022]**

(A) I^- (B) SO_3^{2-} (C) S^{2-} (D) NO_2^- **SA0187**

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EXERCISE # J-ADVANCED

1. Which of the following statement(s) is (are) correct with reference to the ferrous and ferric ions: **[JEE 1998]**

(A) Fe^{3+} gives brown colour with potassium ferricyanide

(B) Fe^{2+} gives blue precipitate with potassium ferricyanide

(C) Fe^{3+} give red colour with potassium thiocyanate

(D) Fe^{2+} gives brown colour with ammonium thiocyanate

SA0138

2. Which of the following statement(s) is /are correct. When a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4? **[JEE 1998]**

(A) A deep red vapours is evolved.

(B) The vapours when passed into NaOH solution gives a yellow solution of Na_2CrO_4

(C) Chlorine gas is evolved

(D) Chromyl chloride is formed.

SA0139

3. An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is a : **[JEE 2000]**

(A) Hg_2^+ salt (B) Cr^{2+} salt (C) Ag^+ salt (D) Pb^{2+} salt

SA0140

4. A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas 'Y'. Identify 'X' and 'Y': **[JEE 2002(Mains)]**

(A) X = CO_2, Y = Cl_2 (B) X = Cl_2, Y = CO_2

(C) X = Cl_2, Y = H_2 (D) X = H_2, Y = Cl_2

SA0141

5. [X] + $H_2SO_4 \rightarrow$ [Y] a colourless gas with irritating smell **[JEE 2003]**

[Y] + $K_2Cr_2O_7$ + $H_2SO_4 \longrightarrow$ green solution

[X] and [Y] are:

(A) SO_3^{2-}, SO_2 (B) Cl^-, HCl (C) S^{2-}, H_2S (D) CO_3^{2-}, CO_2

SA0142

6. A sodium salt of an unknown anion when treated with $MgCl_2$ give white precipitate only on boiling. The anion is: **[JEE 2004]**

(A) SO_4^{2-} (B) HCO_3^- (C) CO_3^{2-} (D) NO_3^-

SA0143

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7. $(NH_4)_2Cr_2O_7$ on heating gives a gas which is also given by: **[JEE 2004]**

(A) heating NH_4NO_2 (B) heating NH_4NO_3

(C) $Mg_3N_2 + H_2O$ (D) $NaNO_2 + H_2O_2$

SA0144

8. A metal nitrate reacts with KI to give a black precipitate which on addition of excess of KI convert into orange colour solution. The cation of metal nitrate is: **[JEE 2005]**

(A) Hg^{2+} (B) Bi^{3+} (C) Pb^{2+} (D) Cu^{+}

SA0145

9. A solution when diluted with H_2O and boiled, it gives a white precipitate. On addition of excess NH_4Cl / NH_4OH, the volume of precipitate decreases leaving behind a white gelatinous precipitate. Identify the precipitate which dissolves in NH_4OH / NH_4Cl. **[JEE 2006]**

(A) $Zn(OH)_2$ (B) $Al(OH)_3$ (C) $Mg(OH)_2$ (D) $Ca(OH)_2$

SA0146

10. $CuSO_4$ decolourises on addition of excess KCN, the product is: **[JEE 2006]**

(A) $[Cu(CN)_4]^{2-}$ (B) Cu^{2+} get reduced to form $[Cu(CN)_4]^{3-}$

(C) $Cu(CN)_2$ (D) CuCN

SA0147

11. Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenylamine as indicator. The number of moles of Mohr's salt required per mole of dichromate is: **[JEE 2007]**

(A) 3 (B) 4 (C) 5 (D) 6

SA0148

12. The species present in solution when CO_2 is dissolved in water are **[JEE 2007]**

(A) $CO_2, H_2CO_3, HCO_3^-, CO_3^{2-}$ (B) H_2CO_3, CO_3^{2-}

(C) CO_3^{2-}, HCO_3^- (D) CO_2, H_2CO_3

SA0149

13. Sodium fusion extract, obtained from aniline, on treatment with iron (II) sulphate and H_2SO_4 in presence of air gives a Prussian blue precipitate. The blue colour is due to the formation of : **[JEE 2007]**

(A) $Fe_4[Fe(CN)_6]_3$ (B) $Fe_3[Fe(CN)_6]_2$

(C) $Fe_4[Fe(CN)_6]_2$ (D) $Fe_3[Fe(CN)_6]_3$

SA0150

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14. **[JEE 2007]**

Column I	Column II
(A) $O_2^- \rightarrow O_2 + O_2^{2-}$	(P) Redox reaction
(B) $CrO_4^{2-} + H^+ \rightarrow$	(Q) One of the products has trigonal planar structure
(C) $MnO_4^- + NO_2^- + H^+ \rightarrow$	(R) Dimeric bridged tetrahedral metal ion
(D) $NO_3^- + H_2SO_4 + Fe^{2+} \rightarrow$	(S) Disproportionation

SA0151

15. A solution of a metal ion when treated with KI gives a red precipitate which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt (II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is **[JEE 2007]**

(A) Pb^{2+} (B) Hg^{2+} (C) Cu^{2+} (D) Co^{2+}

SA0152

16. A solution of colourless salt H on boiling with excess NaOH produces a non-flammable gas. The gas evolution ceases after sometime. Upon addition of Zn dust to the same solution, the gas evolution restarts. The colourless salt(s) H is (are) **[JEE 2008]**

(A) NH_4NO_3 (B) NH_4NO_2 (C) NH_4Cl (D) $(NH_4)_2SO_4$

SA0153

Paragraph for Question Nos. 17 to 19

p-Amino-N, N-dimethylaniline is added to a strongly acidic solution of **X**. The resulting solution is treated with a few drops of aqueous solution of **Y** to yield blue coloration due to the formation of methylene blue. Treatment of the aqueous solution of **Y** with the reagent potassium hexacyanoferrate(II) leads to the formation of an intense blue precipitate. The precipitate dissolves on excess addition of the reagent. Similarly, treatment of the solution of **Y** with the solution of potassium hexacyanoferrate(III) leads to a brown coloration due to the formation of **Z**. **[JEE 2009]**

17. The compound **X** is

(A) $NaNO_3$ (B) NaCl (C) Na_2SO_4 (D) Na_2S

SA0154

18. The compound **Y** is

(A) $MgCl_2$ (B) $FeCl_2$ (C) $FeCl_3$ (D) $ZnCl_2$

SA0154

19. The compound Z is

(A) $Mg_2[Fe(CN)_6]$ (B) $Fe[Fe(CN)_6]$ (C) $Fe_4[Fe(CN)_6]_3$ (D) $K_2Zn_3[Fe(CN)_6]_2$

SA0154

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem Module\s-Salt Anal-Heating Effs,d-Block-Ron of Salt Anal-(OC)\Eng\01_SA\01_ThEx.p65

20. Match each of the reactions given in Column I with the corresponding product(s) given in Column II.

Column I	**Column II**
(A) Cu + dil. HNO_3	(P) NO
(B) Cu + conc. HNO_3	(Q) NO_2
(C) Zn + dil. HNO_3	(R) N_2O
(D) Zn + conc. HNO_3	(S) $Cu(NO_3)_2$
	(T) $Zn(NO_3)_2$

[JEE 2009]

SA0155

21. Passing H_2S gas into a mixture of Mn^{2+}, Ni^{2+}, Cu^{2+} and Hg^{2+} ions in an acidified aqueous solution precipitates **[JEE 2011]**

(A) CuS and HgS (B) MnS and CuS (C) MnS and NiS (D) NiS and HgS

SA0156

22. Reduction of the metal centre in aqueous permanganate ion involves - **[JEE 2011]**

(A) 3 electrons in neutral medium (B) 5 electrons in neutral medium

(C) 3 electrons in weak alkaline medium (D) 5 electrons in acidic medium

SA0157

23. The equilibrium **[JEE 2011]**

$$2Cu^{I} \rightleftharpoons Cu^{o} + Cu^{II}$$

in aqueous medium at 25°C shifts towards the left in the presence of

(A) NO_3^- (B) Cl^- (C) SCN^- (D) CN^-

SA0158

Paragraph for Questions Nos. 24 to 26

When a metal rod M is dipped into an aqueous colourless concentrated solution of compound N, the solution turns light blue. Addition of aqueous NaCl to the blue solution gives a white precipitate O. Addition of aqueous NH_3 dissolves O and gives are intense blue solution. **[JEE 2011]**

24. The metal rod M is -

(A) Fe (B) Cu (C) Ni (D) Co

SA0159

25. The compound N is -

(A) $AgNO_3$ (B) $Zn(NO_3)_2$ (C) $Al(NO_3)_3$ (D) $Pb(NO_3)_2$

SA0159

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26. The final solution contains -

(A) $[Pb(NH_3)_4]^{2+}$ and $[CoCl_4]^{2-}$ (B) $[Al(NH_3)_4]^{3+}$ and $[Cu(NH_3)_4]^{2+}$

(C) $[Ag(NH_3)_2]^{+}$ and $[Cu(NH_3)_4]^{2+}$ (D) $[Ag(NH_3)_2]^{+}$ and $[Ni(NH_3)_6]^{2+}$

SA0159

27. Which of the following hydrogen halides react(s) with $AgNO_3(aq)$ to give a precipitate that dissolves in $Na_2S_2O_3(aq)$: **[JEE 2012]**

(A) HCl (B) HF (C) HBr (D) HI

SA0160

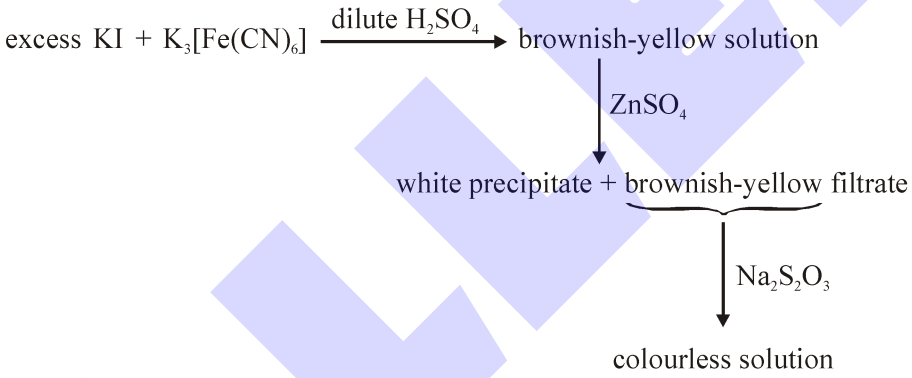
28. The reaction of white phosphorus with aqueous NaOH gives phosphine along with another phosphorus containing compound. The reaction type ; the oxidation states of phosphorus in phosphine and the other product are respectively **[JEE 2012]**

(A) redox reaction ; –3 and –5 (B) redox reaction ; +3 and +5

(C) disproportionation reaction ; –3 and +1 (D) disproportionation reaction ; –3 and +3

SA0161

29. For the given aqueous reactions, which of the statement(s) is (are) true ? **[JEE 2012]**



(A) The first reaction is a redox reaction.

(B) White precipitate is $Zn_3[Fe(CN)_6]_2$.

(C) Addition of filtrate to starch solution gives blue colour.

(D) White precipitate is soluble in NaOH solution.

SA0162

30. Upon treatment with ammonical H_2S , the metal ion that precipitates as a sulfide is -

(A) Fe(III) (B) Al(III) (C) Mg(II) (D) Zn (II) **[JEE 2013]**

SA0163

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Paragraph for Question 31 and 32

An aqueous solution of a mixture of two inorganic salts, when treated with dilute HCl, gave a precipitate **(P)** and a filtrate **(Q)**. The precipitate **(P)** was found to dissolve in hot water. The filtrate **(Q)** remained unchanged, when treated with H_2S in a dilute mineral acid medium. However, it gave a precipitate **(R)** with H_2S in an ammoniacal medium. The precipitate R gave a coloured solution **(S)**, when treated with H_2O_2 in an aqueous NaOH medium. **[JEE 2013]**

31. The coloured solution **(S)** contains

(A) $Fe_2(SO_4)_3$ (B) $CuSO_4$ (C) $ZnSO_4$ (D) Na_2CrO_4

SA0164

32. The precipitate **(P)** contains

(A) Pb^{2+} (B) Hg_2^{2+} (C) Ag^+ (D) Hg^{2+}

SA0164

33. Consider the following list of reagents : **[JEE Adv. 2014]**

Acidified $K_2Cr_2O_7$, alkaline $KMnO_4$, $CuSO_4$, H_2O_2, Cl_2, O_3, $FeCl_3$, HNO_3 and $Na_2S_2O_3$.

The total number of reagents that can oxidise aqueous iodide to iodine is

SA0165

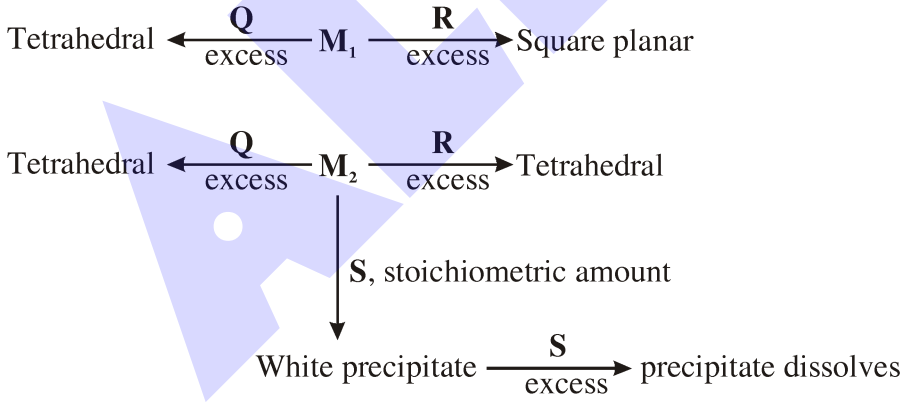
34. Among PbS, CuS, HgS, MnS, Ag_2S, NiS, CoS, Bi_2S_3, and SnS_2 the total number of **BLACK** coloured sulphides is **[JEE Adv. 2014]**

SA0166

Paragraph for Q.No. 35 and 36

An aqueous solution of metal ion $\mathbf{M_1}$ reacts separately with reagents **Q** and **R** in excess to give tetrahedral and square planar complexes, respectively. An aqueous solution of another metal ion $\mathbf{M_2}$ always forms tetrahedral complexes with these reagents. Aqueous solution of $\mathbf{M_2}$ on reaction with reagent **S** gives white precipitate which dissolves in excess of **S**. The reactions are summarized in the scheme given below. **[JEE Adv. 2014]**

SCHEME:



35. $\mathbf{M_1}$, **Q** and **R**, respectively are

(A) Zn^{2+}, KCN and HCl (B) Ni^{2+}, HCl and KCN

(C) Cd^{2+}, KCN and HCl (D) Co^{2+}, HCl and KCN

SA0167

36. Reagent **S** is

(A) $K_4[Fe(CN)_6]$ (B) Na_2HPO_4 (C) K_2CrO_4 (D) KOH

SA0167

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37. Fe^{3+} is reduced to Fe^{2+} by using - **[JEE Adv. 2015]**

(A) H_2O_2 in presence of NaOH (B) Na_2O_2 in water

(C) H_2O_2 in presence of H_2SO_4 (D) Na_2O_2 in presence of H_2SO_4

SA0168

38. The pair(s) of ions where BOTH the ions are precipitated upon passing H_2S gas in presence of dilute HCl, is(are) **[JEE Adv. 2015]**

(A) Ba^{2+}, Zn^{2+} (B) Bi^{3+}, Fe^{3+} (C) Cu^{2+}, Pb^{2+} (D) Hg^{2+}, Bi^{3+}

SA0169

39. The reagent(s) that can selectively precipiate S^{2-} from a mixture of S^{2-} and SO_4^{2-} in aqueous soltuion is(are) : **[JEE(Adv.)-2016]**

(A) $CuCl_2$ (B) $BaCl_2$ (C) $Pb(OOCCH_3)_2$ (D) $Na_2[Fe(CN)_5NO]$

SA0170

40. In the following reaction sequence in aqueous soluiton, the species **X**, **Y** and **Z** respectively, are - **[JEE(Adv.)-2016]**

$$S_2O_3^{2-} \xrightarrow{Ag^+} \underset{\text{clear solution}}{X} \xrightarrow{Ag^+} \underset{\text{white precipitate}}{Y} \xrightarrow{\text{With time}} \underset{\text{black precipitate}}{Z}$$

(A) $[Ag(S_2O_3)_2]^{3-}$, $Ag_2S_2O_3$, Ag_2S (B) $[Ag(S_2O_3)_3]^{5-}$, Ag_2SO_3, Ag_2S

(C) $[Ag(SO_3)_2]^{3-}$, $Ag_2S_2O_3$, Ag (D) $[Ag(SO_3)_3]^{3-}$, Ag_2SO_4, Ag

SA0171

41. Which of the following combination will produce H_2 gas ? **[JEE(Adv.)-2017]**

(A) Zn metal and NaOH(aq) (B) Au metal and NaCN(aq) in the presence of air

(C) Cu metal and conc. HNO_3 (D) Fe metal and conc. HNO_3

SA0172

42. Addition of excess aqueous ammonia to a pink coloured aqueous solution of $MCl_2 . 6H_2O$ (X) and NH_4Cl gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as 1 : 3 electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 B.M., whereas it is zero for complex Y. **JEE(Adv.)-2017]**

Among the following options, which statements is(are) correct ?

(A) The hybridization of the central metal ion in Y is d^2sp^3

(B) Z is tetrahedral complex

(C) Addition of silver nitrate to Y gives only two equivalents of silver chloride

(D) When X and Z are in equilibrium at 0°C, the colour of the solution is pink

SA0173

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal+Heating Effs,d-Block+Ion of Salt Anal\(OC)\Eng\01_SA\01_ThEx.p65

43. The correct option(s) to distinguish nitrate salts of Mn^{2+} and Cu^{2+} taken separately is (are) :-

(A) Mn^{2+} shows the characteristic green colour in the flame test **JEE(Adv.)-2018]**

(B) Only Cu^{2+} shows the formation of precipitate by passing H_2S in acidic medium

(C) Only Mn^{2+} shows the formation of precipitate by passing H_2S in faintly basic medium

(D) Cu^{2+}/Cu has higher reduction potential than Mn^{2+}/Mn (measured under similar conditions)

SA0174

44. The green colour produced in the borax bead test of a chromium(III) salt is due to-

[JEE(Adv.)-2019]

(1) $Cr(BO_2)_3$ (2) CrB (3) $Cr_2(B_4O_7)_3$ (4) Cr_2O_3

SA0175

45. A colorless aqueous solution contains nitrates of two metals, **X** and **Y**. When it was added to an aqueous solution of NaCl, a white precipitate was formed. This precipitate was found to be partly soluble in hot water to give a residue **P** and a solution **Q**. The residue P was soluble in aq. NH_3 and also in excess sodium thiosulfate. The hot solution **Q** gave a yellow precipitate with KI. The metals **X** and **Y**, respectively, are : **[JEE ADV. 2020]**

(A) Ag and Pb (B) Ag and Cd (C) Cd and Pb (D) Cd and Zn

SA0180

46. A mixture of two salts is used to prepare a solution **S**, which gives the following results : **[JEE ADV. 2021]**

White precipitate (s) only $\xleftarrow[\text{Room temperature}]{\text{Dilute NaOH (aq.)}}$ **S** (aq. solution of the salts) $\xrightarrow[\text{Room temperature}]{\text{Dilute HCl (aq.)}}$ White precipitate (s) only

The correct option(s) for the salt mixture is(are)

(A) $Pb(NO_3)_2$ and $Zn(NO_3)_2$ (B) $Pb(NO_3)_2$ and $Bi(NO_3)_3$

(C) $AgNO_3$ and $Bi(NO_3)_3$ (D) $Pb(NO_3)_2$ and $Hg(NO_3)_2$

SA0181

Paragraph for Q.No. 47 and 48

The reaction of $K_3[Fe(CN)_6]$ with freshly prepared $FeSO_4$ solution produces a dark blue precipitate called Turnbull's blue. Reaction of $K_4[Fe(CN)_6]$ with the $FeSO_4$ solution in complete absence of air produces a white precipitate **X**, which turns blue in air. Mixing the $FeSO_4$ solution with $NaNO_3$, followed by a slow addition of concentrated H_2SO_4 through the side of the test tube produces a brown ring. **[JEE ADV. 2021]**

47. Precipitate X is

(A) $Fe_4[Fe(CN)_6]_3$ (B) $Fe[Fe(CN)_6]$

(C) $K_2Fe[Fe(CN)_6]$ (D) $KFe[Fe(CN)_6]$

SA0182

48. Among the following, the brown ring is due to the formation of

(A) $[Fe(NO)_2(SO_4)_2]^{2-}$ (B) $[Fe(NO)_2(H_2O)_4]^{3+}$

(C) $[Fe(NO)_4(SO_4)_2]$ (D) $[Fe(NO)(H_2O)_5]^{2+}$

SA0183

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\6-Salt Anal+Heating Effs,d-Block-Ion of Salt Anal\(OC)\Eng\01_SA\01_TheEx.p65

49. Thermal decomposition of $AgNO_3$ produces two paramagnetic gases. The total number of electrons present in the antibonding molecular orbitals of the gas that has the higher number of unpaired electrons is _____. **[JEE Adv. 2022]**

SA0188

50. The reaction $Pb(NO_3)_2$ and NaCl in water produces a precipitate that dissolves upon the addition of HCl of appropriate concentration. The dissolution of the precipitate is due to the formation of

[JEE Adv. 2022]

(A) $PbCl_2$ (B) $PbCl_4$ (C) $[PbCl_4]^{2-}$ (D) $[PbCl_6]^{2-}$ **SA0189**

51. The treatment of an aqueous solution of 3.74 g of $Cu(NO_3)_2$ with excess KI results in a brown solution along with the formation of a precipitate. Passing H_2S through this brown solution gives another precipitate X. The amount of X (in g) is _____. **[JEE Adv. 2022]**

[Given : Atomic mass of H = 1, N = 14, O = 16, S = 32, K = 39, Cu = 63, I = 127] **SA0190**

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ANSWER KEY

EXERCISE # I

Que.	1	2	3	4	5	6	7			
Ans.	A	C	C	D	D	A,B	A→RS ; B→Q ; C→PQST ; D→PQST			
Que.	8	9	10	11	12	13	14	15	16	17
Ans.	1	D	D	C	D	B	C	B	A	C
Que.	18	19	20	21	22	23	24	25	26	27
Ans.	B	B	D	C	A	B	B	D	B	C
Que.	28	29	30	31	32	33	34	35	36	37
Ans.	A,B	A	C	2	4	2	D	B	D	C
Que.	38	39	40	41	42	43				
Ans.	C	C,D	B	B	D	D				

EXERCISE # II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	C	A	B	B	B	B	B	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	ABCD	C	C	C	B	B	B	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	B	D	C	B	C	C	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	B	D	D	C	B	D	C	D	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	ACD	C	A	ABC	BCD	ACD	BD	D	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	D	D	AC	D	B	C	B	AB	D	B
Que.	61	62	63	64	65	66	67	68		
Ans.	B	B	A	A	C	B	C	D		
Que.	69					70				
Ans.	A→QR; B→Q; C→P; D→ST					A→P; B→S; C→R; D→QR				
Que.	71	72	73	74	75					
Ans.	B	D	C	3	2					

EXERCISE # JEE MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	4	2	2	2	1	2	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	3	4	3	1	4	2	3	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	4	3	1	4	3	4	2	3	2
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	4	2	4	2	4	3	2	C	A	A
Que.	41									
Ans.	B									

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EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABC	ABD	D	C	A	B	A	B	A	B
Que.	11	12	13	14					15	16
Ans.	D	A	A	A→PS ; B→R; C→PQ; D→P					B	AB
Que.	17	18	19	20					21	22
Ans.	D	C	B	A→PS ; B→QS; C→RT; D→QT					A	ACD
Que.	23	24	25	26	27	28	29	30	31	32
Ans.	BCD	B	A	C	ACD	C	ACD	D	D	A
Que.	33	34	35	36	37	38	39	40	41	42
Ans.	7	6 / 7	B	D	AB	CD	A or AC	A	A	ABD
Que.	43	44	45	46	47	48	49	50	51	
Ans.	BD	1	A	AB	C	D	6	C	0.32	

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